www.tntextbooks.in

The epidermis is the outermost layer and bears unicellular root hairs. The cortex is divided into outer wide parenchymatous and inner narrow sclerenchymatous layer. The stele is simple and possesses a central core of xylem in diarch condition with phloem on either side of it (Figure 2.32).

Rhizome

The rhizome in transverse section shows a single layered epidermis covered by cuticle. Some epidermal cells bear multicellular hairs. The Epidermis is followed by two to three layered hypodermis made up of sclerenchyma tissue. A parenchymatous ground tissue is present. The young rhizomes have amphiphloic siphonostele. The older rhizomes have solenostele or dictyostele (Figure 2.33).

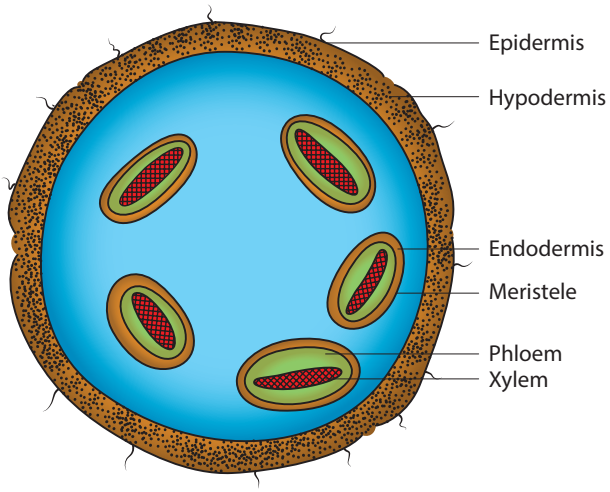


Figure 2.33: T.S. of Rhizome

Petiole

The petiole in T.S. shows a single layered epidermis with thick cuticle. Epidermis is followed by a sclerenchymatous hypodermis which provides mechanical support. There is an extensive parenchymatous ground tissue. The central region possesses a single large horse shoe shaped stele. Xylem forms central core surrounded by phloem (Figure 2.34).

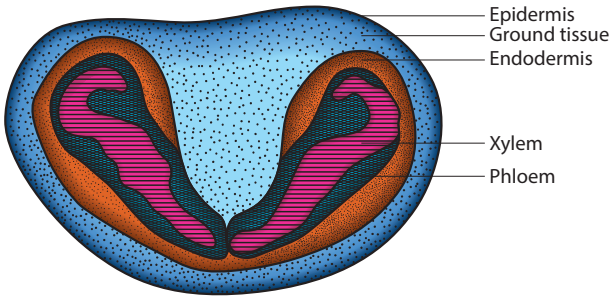


Figure 2.34: T.S. of Petiole

Pinnule

The Pinnule shows upper and lower epidermis. The cells contain chloroplasts. Stomata are confined to lower epidermis. The mesophyll is not differentiated into palisade and spongy parenchyma. The vascular bundle is surrounded by sclerenchymatous bundle sheath.

Reproduction

Adiantum is homosporous. The reproduction takes place by the production of spores. The spores are produced in sporangia. A group of sporangia forms sori. The sori are marginal but the reflex margins of the pinna form a protective membranous structure called *false indusium* (Figure 2.35). The development of sporangium is of leptosporangiate type.

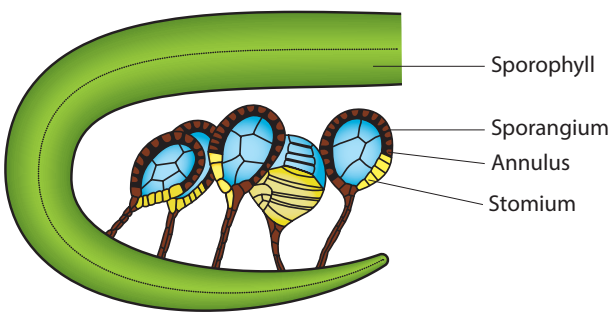


Figure 2.35: V.S. of Sporophyll

The sorus does not show any definite sequence hence fall under mixed type.